AMERICAN UNIVERSITY OF PHNOM PENH STUDY LOCALLY. LIVE GLOBALLY.	American University of Phnom Penh
	School of Digital Technologies
	ITM-390: Machine Learning

Research Proposal

Project Information			
Project title	English	Email Spam Detection System	
	Khmer	ប្រព័ន្ធអង្កេតសារឥតបានការ	
Advisor	Prof. PIN Kuntha	Department	Digital Technology
Team Members			
Team Leader	Menh Chanheng	Student ID	2023445
Member 1	Sorith Pichetroth	Student ID	2024089
Member 2	Heng Shito	Student ID	2023210

I. Introduction

Spam email is a noticeable cybersecurity challenge in modern email communication. Spam emails consume storage, and bandwidth which waste our resources. Moreover, Spam emails can also serve as methods for phishing attacks, malware distribution, and most importantly, financial fraud. Attackers often use email platforms to steal sensitive information such as personal credentials, and financial information like debit or credit card numbers, which may lead to economic losses.

Top email service providers like Gmail and Outlook implement rule-based filtering systems to detect and classify spam emails from our inbox. However, these systems depend too much on manually defined keyword rules, and predefined patterns. Since spammers instantly modify wording, structure, and delivery techniques. Those rule-based mechanisms become a challenge to maintain and update to catch up with spammers' approaches effectively. The limitations of the traditional rule-based algorithm can be the vulnerability for spammers to take advantages. Based on [3], one of the main disadvantages of the existing systems is that the rule-based filters mostly are static, which they couldn't adapt or learn over time, as it is only limited to identify spams that already fit in its known pattern.

Machine learning provides more dynamic, and elastic solutions to this spam problem. In supervised learning, models using labeled datasets are the inputs to train, and output as the prediction of unseen data to classify all those new spam emails. With this approach, lots of problems including advertisement analysis, object detection, spam emails classification are subsequently applied. Moreover, with input as the historical spam email data, supervised machine learning models can automatically recognize complex spam patterns that are impossible for rule-based systems to classify.

To conclude, this research proposes the project of a Spam Email Detection System using supervised machine learning models. The objective of our research aims to recognize and compare multiple supervised classification models to find an effective, and deployable solution for our spam email filtering model.

Creative idea for this project:

- We will collect multiple datasets from different sources, then we will merge it together to create a massive dataset to feed our models.
- We will implement a batch of different models such as Logistic Regression, Naive Bayes, Random Forest, Stochastic Gradient Descent, Gradient Boosting, and Linear SVM, and we will compare all of the models' performance using Accuracy, Precision, Recall, F1-score, Training time to choose the best model for deployment.
- We will use Hybrid Feature Model by combining TF-IDF text features with Email structural features such as Numbers of URLs, Special character count, HTML tag presence, Uppercase ratio, and Email Length, and then we compare Linear models with tree-based models.
- Cross-Domain generalization Test: we will use a test dataset that is completely different from the training dataset. This test will show the robustness, and generalization of the model, and ensure that the model will not be overfitted.

- We will handle class imbalance using class weights, and Threshold tuning.
- We will ensure the model is lightweight that is easy for inference, and deployment. Our strategy is to measure, and compare model size, memory usage, and inference time, then select the low-resource model for the project demonstration.
- Adversarial Robustness: This method prevents adversarial manipulations as attackers often modify text with symbols for example, free to fr33, or Facebook to Faceb00k in order to bypass traditional filter systems.

II. Research Goals and Objectives

a) **Main goal:** To design and comparatively evaluate multiple supervised machine learning models to accurately classify between spam, and non-spam email, and filter spam email away from inbox.

b) **Specific objectives**:

- Collecting and preprocessing email datasets from different sources and building a clean dataset for training and model evaluation.
- Implementing text preprocessing methods such as tokenization, Lowercasing, Stopword removal, Optional stemming, Emoji normalization, and normalization.
- extracting textual features using TF-IDF representation, including unigram and bigram features.
- Training and implementing multiple supervised classification models such as: Logistic Regression, Naive Bayes classifier, SVM, Random Forest, Gradient Boosting, and Stochastic Gradient Descent.
- Evaluating and comparing models' performance using standard classification metrics such as Accuracy, Precision, Recall, F1-score, and training time to select the most effective, and light weight model to deploy spam email filtering systems.
- Analyzing model robustness against adversarial modified spam emails.

Previous Studies related to our project:

- Current Research and Research Gaps:

  Our current research is aiming to transition from rule-based filtering systems to dynamic Machine Learning (ML), and Deep Learning (DL). The research typically utilizes the dataset to train supervised models like Naive Bayes, and Support Vector Machine (SVM). As recent studies have moved toward Feature Engineering, focusing primarily on “Bag of Words” or TF-IDF to convert text into numerical values. However, while these models achieve high accuracy on static datasets, they often struggle when the adversarial spammer tactics appear (e.g “m0n3y”, “h!gh”) and structure manipulation of email metadata. Despite high accuracy reported in many studies, several critical gaps remain what this project aims to address:

1. **Static Feature Space:** Most current filters cannot update their feature space dynamic, as they focus heavily on the message body while ignoring the email header and structural features (HTML tags, uppercase ratios, and URL counts). Our project addresses this by implementing a Hybrid Feature Model by combining TF-IDF and Email structural features.
2. **Neglect of Time Complexity:** Some projects often prioritizes accuracy and precision without even care about the inference time and memory usage. This mistake makes the model impractical for real-world deployment. Our project will clearly measure training/inference time and model size to ensure a lightweight solution. As in [1] suggests that we should consider time complexity as an evaluation matrix since almost all researchers presented their results based on accuracy, recall, precision, and more instead.
3. **Adversarial Vulnerability:** There is a lack of robust methods for dealing with evasion manipulation (e.g., character substitution like "t0y"). Most models are trained on "clean" datasets and failed in real-world scenarios where spammers intentionally obfuscate text. [3] mentions that spammers plan their strategies using evasion, social engineering tactics, and new dictionaries for evading detection. Our research introduces an adversarial robustness to bridge this gap.
4. **Data Generalization:** Many researchers use single-source, manual datasets that perform poorly on real-life data. By merging multiple datasets and performing a Cross-Domain Generalization Test, our project aims to overcome the "overfitting" in current academic ML models.
5. **Class Imbalance:** In real-world inboxes, "ham" or legit emails mostly outweighs spam. Many existing studies use perfectly balanced datasets (50/50) which don't reflect reality. In fact, we will address this through Threshold Tuning and Class Weighting.

III. Methodology

These are all methodologies that our projects will focus on:

1. **Dataset Collection and Preparation:** Our project will utilize multiple spam email datasets available from different sources to increase dataset diversity and capability. The datasets will include labeled spam and non-spam (ham) email.

After collection, all of those datasets will be merged into a unified dataset which we will use for training and testing our model.
These are the following preprocessing steps that will be applied during preparation:

a. Removal of duplicate email's sample
b. Handling missing values
c. Label standardization (spam = 1, ham = 0)
d. Dataset balancing (In case if there're any class imbalance is detected)

The final dataset will be split into:

- Training set: 80%
- Testing set: 20%

2. **Text Preprocessing:** Before we do the feature extraction, raw email texts will follow Natural Language Processing (NLP) preprocessing steps:
    - Tokenization
    - Lowercasing
    - Removal of punctuation
    - Stopword removal
    - Emoji normalization
    - Stemming (To reduce words to their root/dictionary form)
3. **Feature Extraction:** Text data will be converted into numerical representation using Term Frequency-Inverse Document Frequency (TF-IDF). Its representation assigns higher weight to important/rare words while reducing influence of common words. This method could help find words that appear often in one email but not others, as it's quite useful for a model when it tries to spot spam messages as mentioned in [2]. Unigram and bigram features will be considered with the maximum feature size to maintain high efficiency. As a result, we combine it with Email structural features such as Numbers of URLs, Special character count, HTML tag presence, Uppercase ratio, and Email Length as well for better performance.
4. **Model Implementation:** This project will compare both linear and tree-based supervised classification models:
    a. Linear Model:
        - Logistic Regression
        - Support Vector Machine (Linear SVM)
        - Naive Bayes classifier
    b. Tree-based Models:
        - Random Forest
        - Gradient Boosting
5. **Model Evaluation:** Our model performance will be evaluated using the following matrics:
    - Accuracy
    - Precision
    - Recall
    - F1-score
    - Confusion Matrix
    - Training time

6. **Adversarial Robustness Testing:** In order to evaluate real-world reliability, negative modifications will be applied to spam emails in the test set. Modifications include:
    - Character substitution (e.g “free” -> “fr33”)
    - Symbol insertion (e.g “token” -> “token$”)
    - Whitespace manipulation (e.g “winner” -> “w i n n e r”)

    Our models will be evaluated on both the original and modified test sets. Performance will be measured to assess strength against adversarial techniques commonly used by spammers [5].
7. **Final Model Selection:** Based on the performance matrics, strength analysis, procedure efficiency, and training time, the most suitable lightweight model will be selected for deployment recommendation throughout the project.

IV. Timeline of Plan

Task	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10
Literature Review & Proposal Finalization	O									
Dataset Collection & Exploratory Analysis	O	O								
Data Cleaning & Feature Extraction		O	O							
Baseline Model Implementation			O	O						
ML Model Training				O	O	O				
Model Evaluation						O	O			
Web Interface Development							O	O		
Final Report Writing & Code Refactoring								O	O	
Final Presentation Preparation										O

Week 1 — Literature Review & Dataset Collection

- Our group conducts research online to find 3-5 literatures to support our machine learning project.
- We will manually collect data from multiple sources.

Week 2 — Data Cleaning & Preparation

- Dataset collection will end around week 2, which then will begin the data cleaning process of removing duplications, or missing data values.
- Standardize labels (spam=1, ham=0)
- Lastly in week 2, we will split the dataset into 80/20, which the majority is for training the machine learning model and the remaining 20 % for testing afterwards.

Week 3-4 — Text Preprocessing & Model Training & Feature Extraction

- This is the time when we begin the process of tokenization, lowercasing, etc.
- Using TF-IDF features, we will begin to build a baseline model, for example: logistic regression.
- After that, training multiple models to increase the chance of high accuracy. Then we will record the progress in the comparison table we drew up for recording.

Week 5-6 — Hybrid Features & Generalization Test To Evaluate

- This is the time when we implement email structural features such as URLs, special characters or HTML tags. This will combine TF-IDF with structural features which leads to a hybrid feature model.
- Then we will evaluate which model has proved to be the most accurate and precise. That model will be chosen.

Week 7-8 — Web Interface Development

- The chosen model will be chosen as the model to deploy on our web interface which will start development during the week 7-8 timeframe. The process in this week involves improving and fixing code structure of the web interface which will then lead us to the final report.

Week 9 — Final Report Writing & Code Refactoring

- We will continue and finish the final report that includes code snippets, tables or graphs for the model for visual aids.
- Next is to finalize the code to make everything run smoothly without issues.

Week 10 — Final Presentation Preparation

- We prepare for the presentation of our project with visual aids of slides and demos to show.

V. Expected Results

What results do you expect from the project?

By the end of this project, our group expects to achieve:

- A clean and largely combined dataset on email created from joining publicly available dataset from Kaggle and manual collection. The cleaned dataset should have duplicates removed, fill missing values, and labels standardized like spam equates 1, and ham equates 0. We also split the dataset into 80/20 to train the majority and use the remaining to test the accuracy.
- A working model which can detect spam emails via the process of supervised learning system to identify which email is spam or ham. The model would be able to perform in a real world filtering spam scenario.
- Improvement for the detection due to the usage of a hybrid feature like combining text features with email structure features. We believe this will help detection of spam emails which are more difficult to identify.
- Improved robustness of the model's detection capability of trickier spam emails which are modified to avoid detection, for example, "Hello World" → "H3ll0 W0rld" or the usage of symbols. There will be a decrease in accuracy due to this, but the best model should still be able to be sufficient in the real world.
- A comparison of model performance will be held to determine the results of each model trained by using a comparison table for logistic regression with elements like: Accuracy, Precision, Training Time, and so on.
- The deployment of one recommended model chosen as both accurate and fast based on its consumption, model size, and speed.

VI. Final Goal of the Research and Development Project

- ☑ ~~Intended for only ML Class~~
- ☐ Aiming for publication in a journal
- ☐ Planning to extend the research into a startup
- ☐ Other: ……

VII. Expected effects of the development project

- Enhanced Cybersecurity: Detects phishing and fraud attempts which often bypass traditional rule-based filters. It protects users from financial loss and credential theft.
- Operational Efficiency: It automatically filters processes to cut down all kinds of digital clustering. Moreover, it saves user time and optimizes server storage and bandwidth.
- Methodological Insight: Machine Learning models are dynamically adaptive to new "zero-day" spam tactics than traditional static Rule-Based systems.

- Real-World Application: It provides functional, low-latency prototype proving that high-accuracy spam detection which is feasible for real-time personal and enterprise use.

VIII. Students' Role and Experience

Outline each student's specific role in the project and their relevant experience.

Student Names	Role in Project	Experience
MENH Chanheng	AI Engineer, Data Engineer	Natural Processing Language, Computer Vision, Data Wrangling.
HENG Shito	Data collector, Model Evaluator	Basic Python Skill, Basic NLP & AI knowledge
Sorith Pichetroth	Model Trainer & Progress Tracker	Basic Python Skill, Basic NLP & AI knowledge

IX. References

Here lists all references that you cite in the proposal by following IEEE style.

[1] N. Ahmed, R. Amin, H. Aldabbas, D. Koundal, B. Alouffi, and T. Shah, "Machine Learning Techniques for Spam Detection in Email and IoT Platforms: Analysis and Research Challenges," *Security and Communication Networks*, vol. 2022, Art. no. 1862888, pp. 1–19, Feb. 2022, doi: 10.1155/2022/1862888.

[2] R. Oruç and E. Senyürek, "Spam Detection With Naïve Bayes And TF-IDF," *IOSR Journal of Computer Engineering (IOSR-JCE)*, vol. 27, no. 4, ser. 2, pp. 47–51, Jul.–Aug. 2025, doi: 10.9790/0661-2704024751.

[3] S. M. Fateen, S. M. Asif, S. Afroz, N. M. Sami, and Tippanna, "Email Spam Detection Using Machine Learning," *International Journal of Scientific Research in Computer Science Engineering and Information Technology*, vol. 11, no. 3, pp. 248–254, May 2025, doi: 10.32628/CSEIT2511315.

[4] W. A. Awad and S. M. ELseuofi, “Machine Learning Methods for Spam E-Mail Classification,” *International Journal of Computer Science and Information Technology*, vol. 3, no. 1, pp. 173–184, Feb. 2011, doi: 10.5121/ijcsit.2011.3112.
[5] Invisible Technologies, “Model robustness explained: methods, testing, and best practice,” *Invisible Blog*, Aug. 22, 2025. Accessed: Feb. 16, 2026.